Clinton Enwerem

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Summary

Robotics Research Engineer and Electrical and Computer Engineering Ph.D. candidate at the University of Maryland, graduating in Fall 2026. Develop demonstration data pipelines, evaluate imitation and reinforcement learning policies, and implement Python/C++ sensing and control software for dexterous manipulation in simulation and on physical, poly-articulated robots.

Skills

Robotics: ROS/ROS 2, ros2_control, MoveIt 2, MoveIt Servo, SocketCAN, VR teleoperation; **Planning & Control:** RRT*, FCL, state estimation, CBF-CLF QPs, MPC; **Learning:** PyTorch, ACT, DAgger, Diffusion Policy, PPO, flow matching; **Simulation:** Isaac Lab/Sim, MuJoCo, Drake; **Software:** Python, C++, Linux, Docker, git, CI; **Robots:** Fairino FR3, xArm7, UR3e, UR5, Robotiq Hand-E, RealHand A7 (Bimanual Arm-Hand System), RealHand L6, LEAP Hand v1, Unitree G1 (simulation).

Work Experience

Graduate Research Assistant | Aug. 2021 – Present
Institute for Systems Research (ISR), University of Maryland | College Park, MD

• Develop grasp-synthesis, state-estimation, motion-planning, and feedback-control methods for dexterous manipulation under contact uncertainty; implement Python/C++ and ROS 2 software for physical arm–hand systems.
• Build simulated manipulation tasks and synchronized tactile, visual, and proprioceptive data pipelines; generate demonstrations, train imitation and reinforcement learning policies, and evaluate task success and failure modes under changing execution conditions.
• Developed VR teleoperation software for an xArm7-RealHand-L6 robotic manipulator with wrist-to-Cartesian-velocity mapping, relative-joint-position hand control, grasp primitives, thumb-abduction control, and hand-state reset; integrated MoveIt Servo with Dimensional OS (DimOS) for real-time collision detection, arm speed modulation, and velocity damping near kinematic boundaries.

Research Intern | Jun. – Aug. 2022
MATRIX Lab, USM at Southern Maryland | California, MD

• Identified a discrete-time Crazyflie model from flight data and implemented a finite-horizon optimal formation controller in Python and Crazyswarm; executed leader–follower formation flights with 10 physical Crazyflie quadrotors under localization uncertainty.

Robotics Engineer | Mar. 2020 – Feb. 2021
Kognitive Robotics | Lagos, Nigeria

• Designed power and sensor–actuator interfaces in Altium, assembled a mobile robot, and implemented and tested its ROS software on physical hardware.

Robotics Trainee | Jan. 2020 – Feb. 2021
Robotics & Artificial Intelligence Nigeria (RAIN) | Ibadan, Nigeria

• Built ROS-based visual-SLAM, state-estimation, and flight-control software for a quadrotor's onboard computer.

Projects

ParcelStow — Temporal Robustness in Imitation Learning (Paper, project page, Code, Data & Models) | *Isaac Lab, PyTorch* 2026
• Authored ParcelStow, an Isaac Lab benchmark analyzing temporal robustness in imitation learning; generated **970K** expert states (**297** demonstrations) and trained three imitation baselines. At 2× execution speed, ACT success fell to **53%** versus **84%** for the expert, with failures concentrated in high-precision insertion.

Grasp Distance Fields — Feedback Grasp Execution (Project, Paper) | *C++, Python, QP* 2026
• Implemented a CBF-CLF quadratic-program controller for feedback grasp execution; simulated experiments lifted **46 of 50** objects, with a median QP solve time of **0.09 ms**.

realhand_l6_ros2 — Dexterous-Hand ROS 2 Software (Code) | *C++, SocketCAN* 2026
• Implemented joint/tactile-state interfaces, contact-gated control, and MoveIt 2 configuration with mock and virtual-CAN CI tests. Verified communication, feedback, tactile acquisition, and trajectory execution on a physical RealHand L6.

Selected Publications

• **C. Enwerem et al.** "Does Imitation Learning Preserve Temporal Robustness in Dexterous Manipulation? An Expert–Learner Comparison Across Task Execution Speeds." *arXiv preprint*, 2026. arXiv:2609.01453, project page, code, data & models.
• **C. Enwerem et al.** "Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality." *Preprint*, 2026. arXiv:2608.00600, project page.
• **C. Enwerem et al.** "Belief-Informed Hybrid Control with Almost-Sure Target-Set Convergence." *Preprint*, 2026. arXiv:2610.07674.
• Additional first-author publications covering control, safe learning, and robotic manipulation: IROS 2026, CDC 2026, CDC 2025, CDC 2024; full publication list on Google Scholar.

Education

Ph.D., Electrical & Computer Engineering, University of Maryland, College Park, MD. Expected Fall 2026. Aug. 2021 – Present
B.Eng., Electrical Engineering, University of Nigeria, Nsukka. Highest Honors. GPA 3.84/4.0. Aug. 2018